

ZHOB, PakistanWMO: **416200**Lat: **31.358N**Lon: **69.464E**Elev: **1417**StdP: **85.42**Time Zone: **5.00 (E05)**Period: **01-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.2	0.4	-18.7	0.9	14.7	-15.9	1.1	15.5	7.4	12.1	5.5	10.0	2.0	180	0.400

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	11.5	38.3	18.7	37.2	18.5	36.1	18.5	24.5	31.5	23.4	30.7	22.6	30.2	3.2	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.7	20.7	29.2	21.4	19.2	28.3	20.4	18.0	27.4	83.4	31.7	78.7	30.9	74.7	30.1	30.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.3	6.4	5.2	-2.9	40.8	1.6	1.0	-4.1	41.5	-5.0	42.1	-6.0	42.6	-7.2	43.3		
(5)			-5.2	26.9	1.5	2.1	-6.2	28.4	-7.1	29.6	-7.9	30.8	-9.0	32.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.1	7.4	9.7	15.1	21.5	26.4	29.8	30.1	29.3	26.7	21.4	14.3	9.1
	DBStd	8.73	2.47	3.73	3.80	3.51	2.96	2.51	2.30	2.23	2.06	3.11	2.94	2.63
	HDD10.0	186	86	46	4	0	0	0	0	0	0	0	3	48
	HDD18.3	1124	340	242	114	12	0	0	0	0	0	8	124	284
	CDD10.0	3878	4	38	161	345	508	594	622	598	501	353	132	21
	CDD18.3	1774	0	0	13	107	250	344	364	340	251	103	3	0
	CDH23.3	21782	0	8	168	1126	3067	4645	4729	4158	2797	1024	57	4
	CDH26.7	11189	0	1	32	425	1583	2649	2641	2180	1337	338	3	0
Wind	WSAvg	2.4	2.4	2.2	2.5	2.7	2.6	2.7	2.5	2.5	2.3	2.1	2.0	2.2
Precipitation	PrecAvg	265	15	23	40	30	12	15	54	43	12	2	6	11
	PrecMax	419	78	72	116	134	58	77	163	151	59	26	80	74
	PrecMin	117	0	0	0	0	0	0	2	0	0	0	0	0
	PrecStd	76	15	19	31	31	12	18	37	37	16	5	13	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.6	24.4	29.7	34.4	37.8	40.0	39.8	38.9	36.7	33.2	27.0	23.1
		MCWB	7.5	9.9	12.3	14.5	17.0	19.3	19.9	20.0	16.1	15.3	11.7	10.1
	2%	DB	17.7	21.6	27.3	32.6	36.4	38.4	38.1	37.2	35.0	31.7	25.1	20.2
		MCWB	6.6	9.2	11.8	14.6	16.2	17.9	19.6	19.6	17.5	14.5	11.0	8.7
	5%	DB	15.8	19.4	25.2	30.9	35.1	37.1	36.8	35.9	33.9	30.1	23.4	18.3
		MCWB	5.9	8.4	11.2	13.9	15.8	17.9	19.8	19.4	17.1	13.9	10.4	7.5
	10%	DB	14.1	17.2	23.0	29.2	33.7	35.8	35.4	34.7	32.7	28.4	21.7	16.4
		MCWB	5.3	7.7	10.5	13.8	15.5	17.8	20.3	19.4	17.3	13.5	9.7	6.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	12.7	15.5	19.9	23.4	24.6	26.6	26.0	24.1	19.7	14.2	11.4
		MCDB	16.0	18.4	23.9	29.7	33.3	33.3	33.9	33.1	31.9	26.6	21.9	21.8
	2%	WB	7.8	10.7	13.8	17.3	21.2	22.8	24.7	24.0	22.0	17.7	13.0	9.5
		MCDB	15.0	17.8	22.4	27.2	31.8	32.8	31.6	30.3	29.6	26.5	20.8	17.9
	5%	WB	6.8	9.6	12.5	15.9	19.3	21.7	23.7	23.1	21.0	16.5	11.8	8.2
		MCDB	14.1	17.2	21.8	25.8	29.3	30.9	30.3	29.6	28.4	26.0	19.8	16.7
	10%	WB	5.9	8.5	11.4	15.1	17.8	21.1	22.7	22.3	20.2	15.5	10.8	7.1
		MCDB	12.8	15.9	20.7	24.9	28.4	30.4	30.2	29.3	27.3	24.9	19.2	15.2
Mean Daily Temperature Range	5% DB	MDBR	12.7	12.1	12.4	12.1	12.5	11.5	11.4	10.9	12.0	13.0	13.2	13.6
		MCDBR	15.8	15.3	15.5	14.2	13.9	12.6	12.6	12.3	13.2	13.8	15.0	16.6
		MCWBR	7.7	6.6	5.5	4.2	3.8	3.8	3.8	4.2	4.5	4.0	5.7	7.8
	5% WB	MCDBR	14.2	13.4	13.7	12.5	12.7	11.5	11.7	11.2	12.2	12.8	13.1	15.4
		MCWBR	7.7	6.7	6.3	5.0	5.8	4.3	4.6	4.3	4.8	4.9	5.8	8.1
Clear-Sky Solar Irradiance	taub		0.329	0.389	0.421	0.470	0.479	0.555	0.635	0.582	0.450	0.393	0.359	0.320
	taud		2.301	2.115	2.035	1.946	1.931	1.772	1.649	1.757	2.020	2.104	2.194	2.338
	Ebn at Noon		888	857	857	828	821	756	695	727	818	839	840	878
	Edh at Noon		106	141	165	187	192	224	252	224	165	141	117	97
All-Sky Solar Radiation	RadAvg		3.62	4.25	5.46	6.52	7.32	7.17	6.68	6.39	6.08	5.22	4.05	3.51
	RadStd		0.26	0.39	0.40	0.34	0.30	0.26	0.24	0.36	0.24	0.15	0.21	0.18